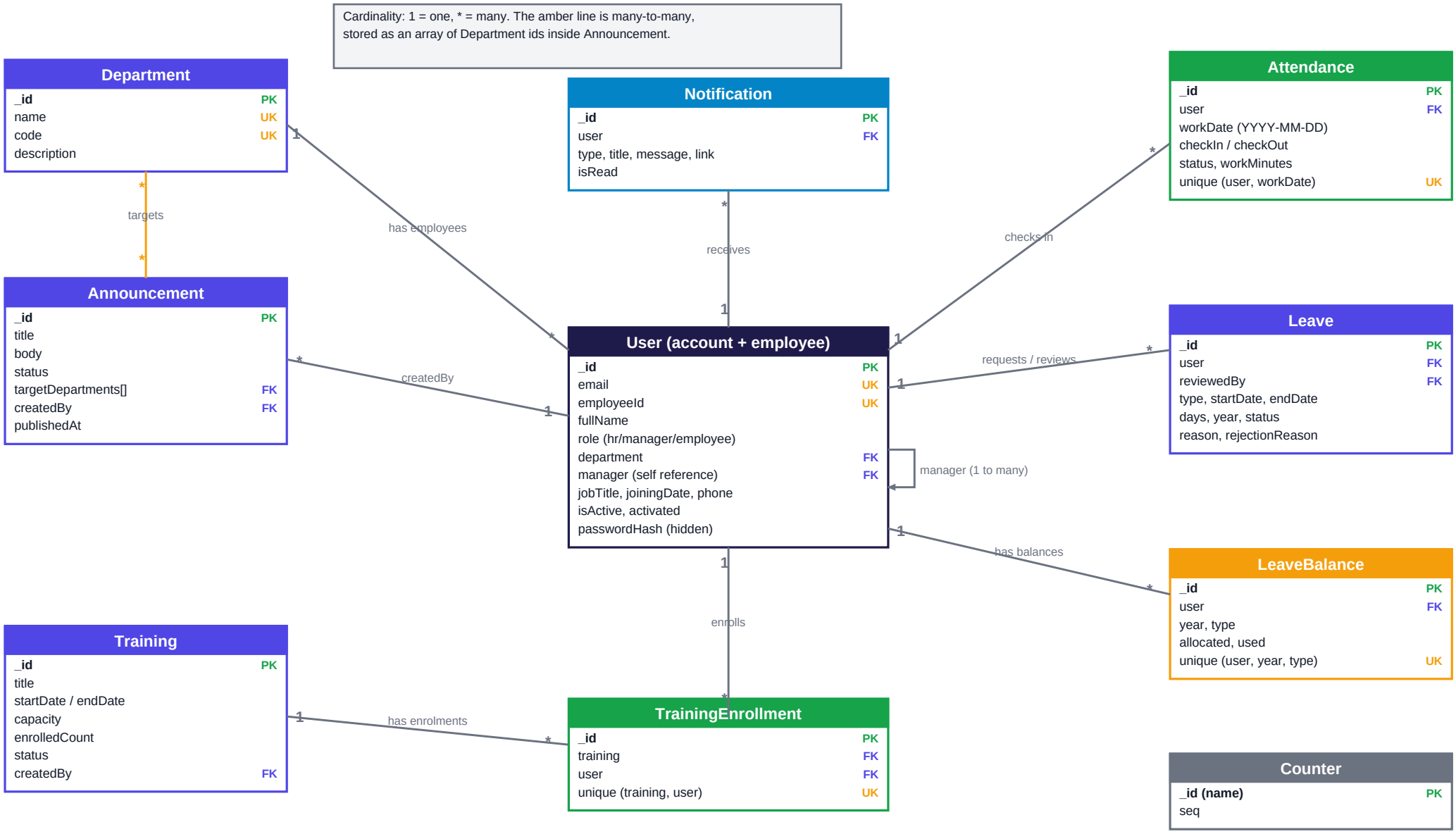
Entity Relationship (ER) Diagram

Ten MongoDB collections and how they reference each other. PK = primary key, FK = reference (ObjectId), UK = unique.



Relationships and design decisions

Every reference in the database, with the field that stores it

From (many side)	To (one side)	Field	Type	Meaning
User	Department	user.department	many to one	An employee belongs to at most one department.
User	User	user.manager	many to one (self)	Reporting line: an employee has one manager, a manager has many team members.
Attendance	User	attendance.user	many to one	One record per user per workday (unique user + workDate).
Leave	User	leave.user	many to one	Who requested the leave.
Leave	User	leave.reviewedBy	many to one	Manager or HR who approved or rejected it.
LeaveBalance	User	leavebalance.user	many to one	Yearly entitlement per leave type (unique user + year + type).
Announcement	Department	announcement.targetDepartments[]	many to many	Empty array means the whole organisation.
Announcement	User	announcement.createdBy	many to one	HR author.
Training	User	training.createdBy	many to one	HR or manager who created the program.
TrainingEnrollment	Training	enrollment.training	many to one	Which program.
TrainingEnrollment	User	enrollment.user	many to one	Which employee (unique training + user: no duplicate enrolments).
Notification	User	notification.user	many to one	Recipient; read or unread.
Counter	-	-	standalone	Auto-increment source for employee IDs (EMP-0001 ...).

Why the design looks like this

- **Users and Employees are one collection.** A login account and an HR profile describe the same person and are always read together. The password hash is hidden (select: false) and removed from JSON output.
- **References, not embedding.** Departments, users, leaves and trainings are queried on their own and keep growing, so they link by ObjectId. Mongoose populate() fetches the linked document; aggregate() with $lookup builds reports inside MongoDB.
- **Dates as text for calendar days.** workDate, leave dates and joining date are YYYY-MM-DD strings in the organisation timezone, so "the 28th" never shifts across timezones. Check-in and check-out are real UTC timestamps.
- **Leave balances are the leave policy.** Defaults (casual 12, sick 10, earned 15) come from configuration; each year a row per user and type is created on demand and HR can change it through the API.
- **Database constraints do the hard work.** Unique indexes stop duplicate emails, employee IDs, daily attendance, enrolments and balance rows even when requests arrive at the same moment.
- **Counters collection** gives safe sequential employee IDs because the increment is a single atomic operation.

Data dictionary

Fields, types and rules for each collection (timestamps createdAt and updatedAt are added automatically)

users

Accounts and employee profiles for all three roles.

Field	Type	Required	Rules / notes
fullName	String	Yes	2 to 80 characters, trimmed
email	String	Yes	Unique, lowercase
employeeId	String	Yes	Unique, uppercase, auto-generated as EMP-0001 if HR leaves it blank
passwordHash	String	No	bcrypt hash; not selected by default; empty until the invite is accepted
role	Enum	Yes	hr, manager or employee (default employee)
department	ObjectId (Department)	No	Reference
manager	ObjectId (User)	No	Must be an active manager or HR; cannot be the employee themself
jobTitle, phone	String	No	Phone is the only contact detail stored
joiningDate	String (YYYY-MM-DD)	Yes	Defaults to the day of creation
employmentStatus	Enum	No	full-time, part-time, contract, intern
isActive	Boolean	No	Deactivated users cannot log in; default true
activated	Boolean	No	True once a password was set; default false
passwordChangedAt	Date	No	Older tokens are rejected
inviteTokenHash / inviteExpires	String / Date	No	SHA-256 of the one-time invite token; valid 7 days; hidden
lastLoginAt	Date	No	Set on each login

Indexes: unique email; unique employeeId; manager; department + isActive; role + isActive; fullName.

departments

Organisation units used for grouping, filtering and announcement targeting.

Field	Type	Required	Rules / notes
name	String	Yes	Unique, 2 to 60 characters
code	String	Yes	Unique, uppercase, up to 10 characters
description	String	No	Up to 300 characters

Indexes: unique name; unique code. A department with employees cannot be deleted.

attendance

One row per person per working day.

Field	Type	Required	Rules / notes
user	ObjectId (User)	Yes	Reference
workDate	String (YYYY-MM-DD)	Yes	Local date in the organisation timezone
checkIn	Date (UTC)	Yes	Time of check-in
checkOut	Date (UTC)	No	Empty until check-out; a past day without it is flagged missed check-out
status	Enum	Yes	present, or late (after work start + grace minutes)
workMinutes	Number	No	Check-out minus check-in

Indexes: unique (user, workDate); workDate.

leaves

Leave requests and their review outcome.

Field	Type	Required	Rules / notes
user	ObjectId (User)	Yes	Requester
type	Enum	Yes	casual, sick, earned
startDate, endDate	String (YYYY-MM-DD)	Yes	End not before start; same calendar year
year	Number	Yes	Year of the start date; selects the balance row
days	Number	Yes	Working days only (weekends and holidays excluded)
reason	String	Yes	5 to 500 characters
status	Enum	Yes	pending, approved, rejected, cancelled
reviewedBy / reviewedAt	ObjectId / Date	No	Set when approved or rejected
rejectionReason	String	No	Required when rejecting

Indexes: user + startDate; status + createdAt; startDate + endDate.

leavebalances

Yearly leave entitlement and usage; doubles as the leave policy.

Field	Type	Required	Rules / notes
user	ObjectId (User)	Yes	Reference
year	Number	Yes	Calendar year
type	Enum	Yes	casual, sick, earned
allocated	Number	Yes	Days granted
used	Number	No	Days consumed by approved leave; never above allocated

Indexes: unique (user, year, type).

announcements

Messages from HR to the whole organisation or selected departments.

Field	Type	Required	Rules / notes
title, body	String	Yes	Title 3 to 120, body 3 to 2000 characters
targetDepartments	[ObjectId (Department)]	No	Empty = everyone
status	Enum	No	draft (default), published, archived
publishedAt	Date	No	Set when published
createdBy	ObjectId (User)	Yes	HR author

Indexes: status + publishedAt.

trainings

Training programs.

Field	Type	Required	Rules / notes
title, description	String	Yes	Title 3 to 120 characters
startDate, endDate	Date	Yes	End not before start
location	String	No	Room or online
capacity	Number	No	0 means unlimited
enrolledCount	Number	No	Kept in step with enrolments by atomic updates
status	Enum	No	draft (default), published, archived
createdBy	ObjectId (User)	Yes	HR or manager

Indexes: status + startDate; createdBy.

trainingenrollments

Which employee joined which program.

Field	Type	Required	Rules / notes
training	ObjectId (Training)	Yes	Reference
user	ObjectId (User)	Yes	Reference

Indexes: unique (training, user); user + createdAt.

notifications

In-app alerts (leave decisions, announcements, new trainings).

Field	Type	Required	Rules / notes
user	ObjectId (User)	Yes	Recipient
type	Enum	Yes	leave, announcement, training, system
title, message, link	String	Title yes	Link opens the related page
isRead	Boolean	No	Default false; drives the unread count

Indexes: user + isRead + createdAt.

counters

Atomic sequence numbers.

Field	Type	Required	Rules / notes
_id	String	Yes	Sequence name, for example employeeId
seq	Number	No	Incremented with a single atomic $inc

Indexes: primary key only.